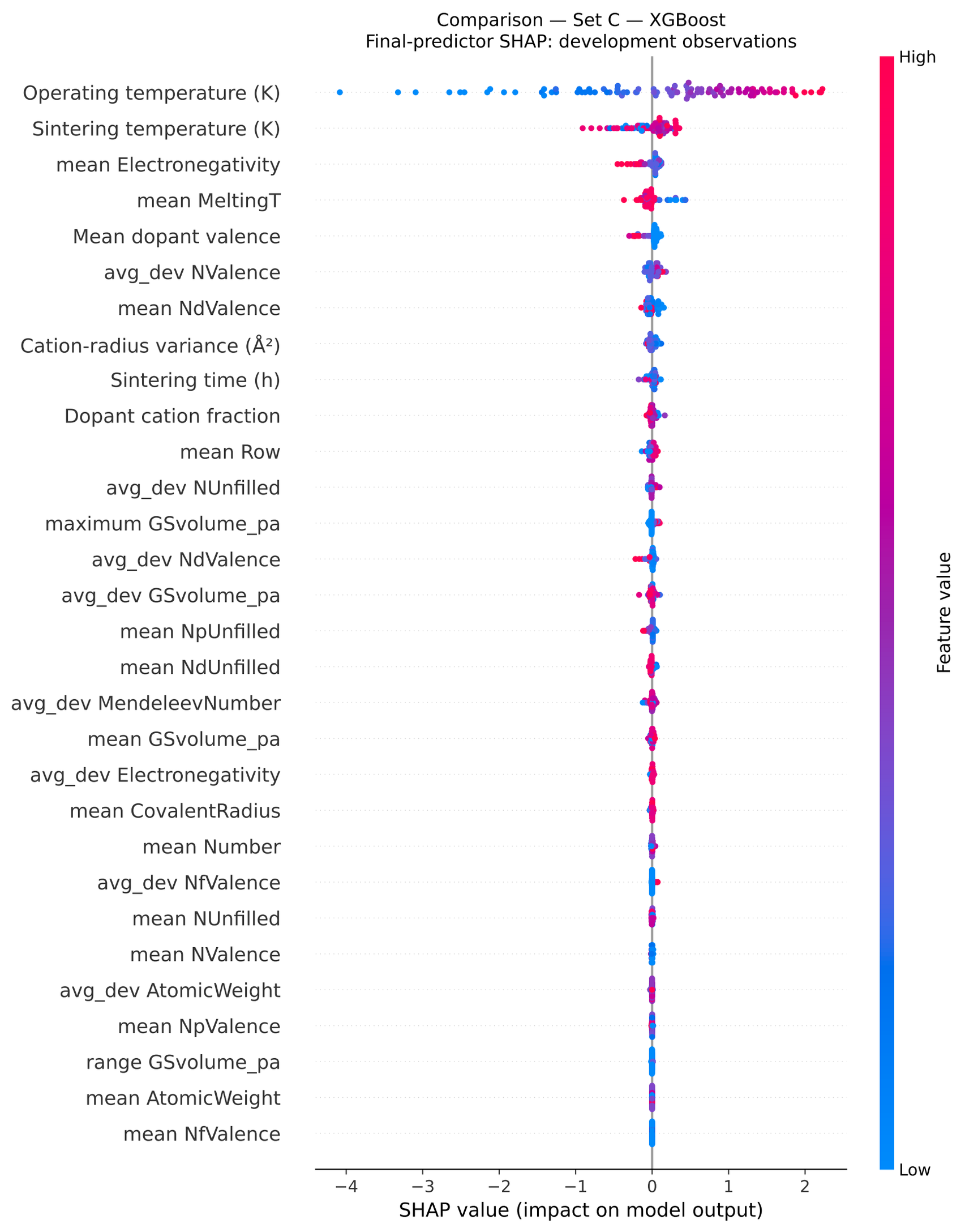

Comparison — Set C — XGBoost
Final-predictor SHAP: development observations
Operating temperature (K)
Sintering temperature (K)
mean Electronegativity
mean MeltingT
Mean dopant valence
avg_dev NValence
mean NdValence
Cation-radius variance (Å²)
Sintering time (h)
Dopant cation fraction
mean Row
avg_dev NUnfilled
maximum GSvolume_pa
avg_dev NdValence
avg_dev GSvolume_pa
mean NpUnfilled
mean NdUnfilled
avg_dev MendeleevNumber
mean GSvolume_pa
avg_dev Electronegativity
mean CovalentRadius
mean Number
avg_dev NfValence
mean NUnfilled
mean NValence
avg_dev AtomicWeight
mean NpValence
range GSvolume_pa
mean AtomicWeight
mean NfValence
−4
−3
−2
−1
0
1
2
SHAP value (impact on model output)
High
Low
Feature value